Student Name: _____

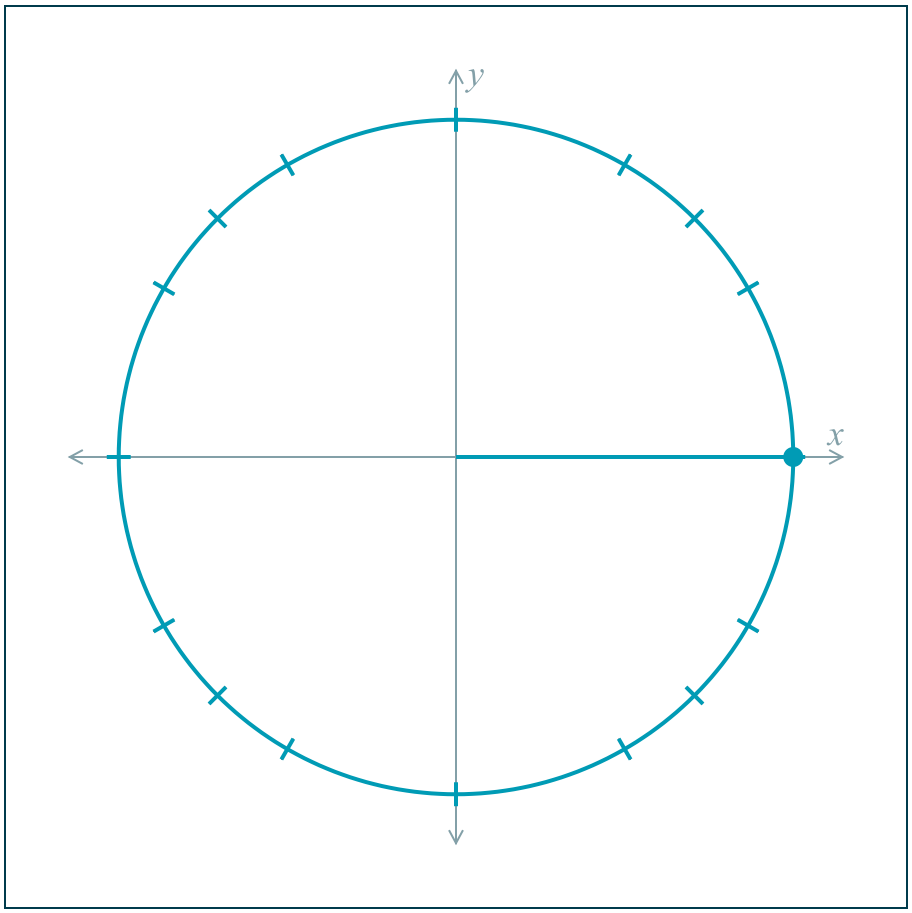
Class Name: **ALEKS Integrated Mathematics II**

Number of Questions: **30**

Instructor Name: **Bodanske, Daniel**

Question 1 of 30

Sketch $\theta = -\frac{4\pi}{3}$ in standard position.



Question 2 of 30

Find the terminal point on the unit circle determined by 0 radians.

Use exact values, not decimal approximations.

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Question 3 of 30

Find the exact value of $\cos 150°$.

Question 4 of 30

Find the exact values below.

$$\cot \frac{3\pi}{2}$$

$$\sec \frac{3\pi}{2}$$

Find the exact values below. If applicable, click on "Undefined".

$\cot \frac{3\pi}{2} =$ _____

$\sec \frac{3\pi}{2} =$ _____

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Question 5 of 30

Let θ be an angle in quadrant III such that $\cos\theta = -\frac{5}{8}$.

Find the exact values of $\csc\theta$ and $\cot\theta$.

Let θ be an angle in quadrant III such that $\cos\theta = -\frac{5}{8}$.

Find the exact values of $\csc\theta$ and $\cot\theta$.

$\csc\theta =$ ☐

$\cot\theta =$ ☐

Question 6 of 30

Let θ be an angle in quadrant IV such that $\tan\theta = -\frac{6}{5}$.

Find the exact values of $\cos\theta$ and $\csc\theta$.

Question 7 of 30

Simplify.

$$\tan^3 x \csc^2 x - \tan^3 x$$

Use algebra and the fundamental trigonometric identities.
Your answer should be a number or use a single trigonometric function.

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Question 8 of 30

Use a cofunction to write an expression equal to $\sec 68°$.

$\sec 68° = \square$

Question 9 of 30

Simplify.

$$\frac{\cot x}{\cos x}$$

Use algebra and the fundamental trigonometric identities.
Your answer should be a number or use a single trigonometric function.

Question 10 of 30

Complete the proof of the identity by choosing the Rule that justifies each step.

$$(1 + \tan x)^2$$
$$= 1 + 2\tan x + \tan^2 x$$
$$= \sec^2 x + 2\tan x$$

Question 11 of 30

Prove the identity.

$$\cos^2 x - \sin^2 x = 2\cos^2 x - 1$$

Question 12 of 30

Find the exact value of $\cos(-15°)$ by using a sum or difference formula.

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Question 13 of 30

Use a sum or difference formula to find the exact value of the following.

$$\frac{\tan\frac{31\pi}{36} - \tan\frac{19\pi}{36}}{1 + \tan\frac{31\pi}{36}\tan\frac{19\pi}{36}}$$

Question 14 of 30

Find $\sin 2x$, $\cos 2x$, and $\tan 2x$ if $\cos x = -\frac{3}{\sqrt{10}}$ and x terminates in quadrant III.

Question 15 of 30

Use a half-angle formula to find the exact value of $\sin 112.5°$.

Question 16 of 30

Answer the following. Give exact values for your answers.

(a) Find an angle between $0°$ and $360°$ that is coterminal with $-120°$.

_____ °

(b) Find an angle between 0 and 2π that is coterminal with $\frac{9\pi}{4}$.

_____ radians

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Question 17 of 30

The maximum height H of an object launched with initial velocity v_0 at an angle θ is given by the following formula.

$$H = \frac{v_0^2 \sin^2\theta}{2g}$$

If $g = 9.8\ \text{m/s}^2$, find the height of an object when $v_0 = 100\ \text{m/s}$ and $\theta = 0.64\ \text{rad}$.

Do not round any intermediate computations. Round your answer to the nearest hundredth.

Question 18 of 30

Find the reference angle for $200°$.

Question 19 of 30

Find all solutions of the equation in the interval $[0, 2\pi)$.

$$2\cos\theta - \sqrt{3} = 0$$

Write your answer in radians in terms of π.
If there is more than one solution, separate them with commas.

Question 20 of 30

Find all solutions of the equation in the interval $[0, 2\pi)$.

$$\tan\theta + 1 = 0$$

Write your answer in radians in terms of π.

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Question 21 of 30

Use the ALEKS calculator to evaluate each expression.
Give your answers in radians.
Round them to the nearest hundredth.

$\tan^{-1}(-2.42) =$ _____

$\cos^{-1}(1.26) =$ _____

$\sin^{-1}(0.81) =$ _____

Question 22 of 30

Find all solutions to the equation.

$$\cos\theta + 1 = 1$$

Write your answer in radians in terms of π.

Example: $\theta = \frac{\pi}{5} + 2k\pi,\ k \in \mathbb{Z}$ or $\theta = \frac{\pi}{7} + k\pi,\ k \in \mathbb{Z}$

Question 23 of 30

Find all solutions to the equation.

$$\sqrt{3}\csc\theta - 2 = 0$$

Write your answer in radians in terms of π.

Example: $\theta = \frac{\pi}{5} + 2k\pi,\ k \in \mathbb{Z}$ or $\theta = \frac{\pi}{7} + k\pi,\ k \in \mathbb{Z}$

Question 24 of 30

Find all solutions of the equation in the interval $[0,\ 2\pi)$.

$$4\cos^2 x - 1 = 0$$

Write your answer in radians in terms of π.

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Question 25 of 30

Find all solutions of the equation in the interval $[0, 2\pi)$.

$$\csc^2 x + 3\csc x + 2 = 0$$

Write your answer in radians in terms of π.
If there is more than one solution, separate them with commas.

Question 26 of 30

Find all solutions of the equation in the interval $[0, 2\pi)$.

$$\tan^2 x + 2\sec x = -1$$

Write your answer in radians in terms of π.
If there is more than one solution, separate them with commas.

Question 27 of 30

Suppose a projectile is fired from a cannon with velocity v_0 and angle of elevation θ. The horizontal distance $R(\theta)$ it travels (in feet) is given by the following.

$$R(\theta) = \frac{(v_0)^2 \sin 2\theta}{32}$$

If $v_0 = 80$ ft/s, what angle θ (in radians) should be used to hit a target on the ground 116 feet in front of the cannon?

Do not round any intermediate computations, and round your answer(s) to the nearest hundredth of a radian. (If there is more than one answer, enter additional answers with the "or" button.)

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Question 28 of 30

Find all solutions to the equation.

$$\cot^2 x - 3 = 0$$

Write your answer in radians in terms of π.

Example: $x = \frac{\pi}{5} + 2k\pi,\ k \in \mathbb{Z}$ or $x = \frac{\pi}{7} + k\pi,\ k \in \mathbb{Z}$

Question 29 of 30

Find all solutions to the equation.

$$2\sec^2\theta - 3\sec\theta - 2 = 0$$

Write your answer in radians in terms of π.

Example: $\theta = \frac{\pi}{5} + 2k\pi,\ k \in \mathbb{Z}$ or $\theta = \frac{\pi}{7} + k\pi,\ k \in \mathbb{Z}$

Question 30 of 30

Find all solutions to the equation.

$$\sin x = -\cos^2 x - 1$$

Write your answer in radians in terms of π.

Example: $x = \frac{\pi}{5} + 2k\pi,\ k \in \mathbb{Z}$ or $x = \frac{\pi}{7} + k\pi,\ k \in \mathbb{Z}$

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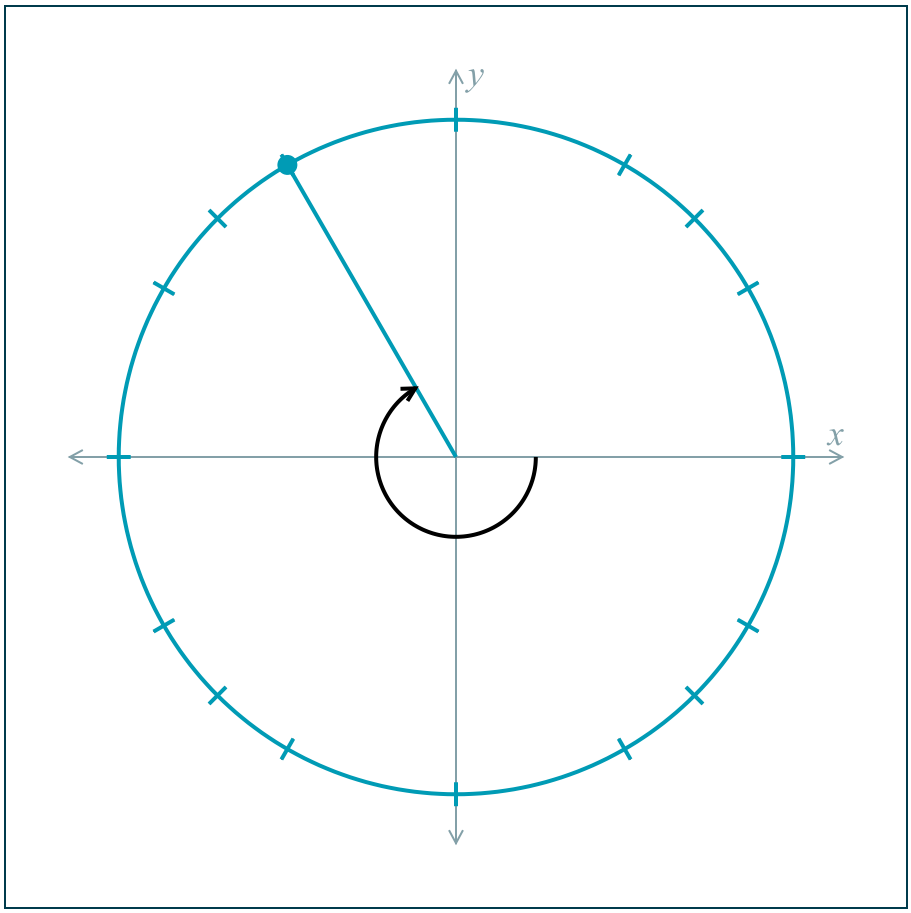
Class Name: **ALEKS Integrated Mathematics II** Number of Questions: **30**

Question 1 of 30



Question 2 of 30

$(x, y) = (1, 0)$

Question 3 of 30

$\cos 150° = -\frac{\sqrt{3}}{2}$

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Question 4 of 30

$\cot \frac{3\pi}{2} = 0$

$\sec \frac{3\pi}{2}$ Undefined

Question 5 of 30

$\csc\theta = -\frac{8\sqrt{39}}{39}$

$\cot\theta = \frac{5\sqrt{39}}{39}$

Question 6 of 30

$\cos\theta = \frac{5\sqrt{61}}{61}$

$\csc\theta = -\frac{\sqrt{61}}{6}$

Question 7 of 30

$\tan x$

Question 8 of 30

$\sec 68° = \csc 22°$

Question 9 of 30

$\csc x$

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Question 10 of 30

Statement	Rule
$(1 + \tan x)^2$	
$= 1 + 2\tan x + \tan^2 x$	Algebra
$= \sec^2 x + 2\tan x$	Pythagorean

Question 11 of 30

Statement	Reason
$\cos^2 x - \sin^2 x$	
$= \cos^2 x - (1 - \cos^2 x)$	Pythagorean
$= \cos^2 x - 1 + \cos^2 x$	Algebra
$= 2\cos^2 x - 1$	Algebra

Question 12 of 30

$$\frac{\sqrt{6} + \sqrt{2}}{4}$$

Question 13 of 30

$$\sqrt{3}$$

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Question 14 of 30

$\sin 2x = \frac{3}{5}$

$\cos 2x = \frac{4}{5}$

$\tan 2x = \frac{3}{4}$

Question 15 of 30

$\sin 112.5° = \frac{\sqrt{2+\sqrt{2}}}{2}$

Question 16 of 30

(a) $240°$

(b) $\frac{\pi}{4}$ radians

Question 17 of 30

$H = 181.96 \text{ m}$

Question 18 of 30

$20°$

Question 19 of 30

$\theta = \frac{\pi}{6}, \frac{11\pi}{6}$

Question 20 of 30

$\theta = \frac{3\pi}{4}, \frac{7\pi}{4}$

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Question 21 of 30

$\tan^{-1}(-2.42) = -1.18$

$\cos^{-1}(1.26) =$ Undefined

$\sin^{-1}(0.81) = 0.94$

Question 22 of 30

$\theta = \frac{\pi}{2} + k\pi,\ k \in \mathbb{Z}$

Question 23 of 30

$\theta = \frac{\pi}{3} + 2k\pi,\ k \in \mathbb{Z}$ or $\theta = \frac{2\pi}{3} + 2k\pi,\ k \in \mathbb{Z}$

Question 24 of 30

$x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$

Question 25 of 30

$x = \frac{3\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$

Question 26 of 30

$x = \frac{2\pi}{3}, \frac{4\pi}{3}$

Question 27 of 30

$\theta = 0.31$ rad

or $\theta = 1.26$ rad

Question 28 of 30

$x = \frac{\pi}{6} + k\pi,\ k \in \mathbb{Z}$

or $x = \frac{5\pi}{6} + k\pi,\ k \in \mathbb{Z}$

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Question 29 of 30

$$\theta = \frac{\pi}{3} + 2k\pi\ ,\ \ k \in \mathbb{Z}$$

or $\theta = \frac{5\pi}{3} + 2k\pi\ ,\ \ k \in \mathbb{Z}$

Question 30 of 30

$x = \frac{3\pi}{2} + 2k\pi,\ k \in \mathbb{Z}$

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